

Date of publication xxxx 00, 0000, date of current version Nov 10, 2025.

Digital Object Identifier 00.0000/ACCESS.2025.0000000

Automated Measurement of Horizontal Gauge Deviation in Railroads Using Depth Sensor Camera and Machine Learning

VAN TRUNG LE¹, HWAPYEONG SONG¹, HUSNU S. NARMAN¹, PINGPING ZHU², AMMAR ALZARRAD³, ABBY CISKO⁴, and JEREMY BEASLEY⁴

¹Department of Computer Science, Marshall University, Huntington, WV 25755, USA (e-mail: le57@marshall.edu; song24@marshall.edu; narman@marshall.edu)

²Department of Biomedical and Electrical Engineering, Marshall University, Huntington, WV 25755, USA (e-mail: zhup@marshall.edu)

³Department of Civil Engineering, Marshall University, Huntington, WV 25755, USA (e-mail: alzarad@marshall.edu)

⁴US Army Engineer Research and Development Center (ERDC), Vicksburg, MS 39180, USA (e-mail: Abby.R.Cisko@erd.c.dren.mil; Thomas.J.Beaesley@erd.c.dren.mil)

Corresponding author: Husnu S. Narman (e-mail: narman@marshall.edu).

This project was partially funded by the U.S. Army Engineer Research and Development Center (ERDC).

ABSTRACT Detecting horizontal gauge deviations on railroad tracks is critical for ensuring operational safety and enabling cost-effective maintenance planning. Traditional approaches rely on expensive multi-sensor systems to achieve high accuracy. This paper proposes a more affordable method using data captured from an Intel D435 Depth Camera mounted on a moving vehicle. The approach integrates machine learning-based segmentation models, specifically YOLOv5x-seg for initial detection and SAMURAI for robust tracking, combined with depth-based 3D reconstruction to measure the distance between railroad tracks. A detailed denoising pipeline mitigates the noise from vegetation, sunlight, and vehicle motion, while geometric processing techniques ensure an accurate pairing of track edges for gauge calculation. Field measurements showed an average error of 1.2%, highlighting the challenges introduced by outdoor variability. Limitations such as limited dataset, vegetation-induced segmentation errors, and the requirement that the camera's viewing direction align with the vehicle's direction of motion are analyzed, along with alternative strategies such as rotation correction. The results show the feasibility of a low-cost, camera-based solution for horizontal gauge detection, while also identifying critical challenges that must be addressed for reliable deployment in real-world environments. The novelty of this work lies in combining a custom depth-denoising method, a modified version of the SAMURAI tracker that can follow both tracks at once using multi-processing, and a 3D gauge measurement technique based on points pairing. These components are integrated into a practical, low-cost system using only a single depth camera, offering a simpler alternative to traditional multi-sensor setups.

INDEX TERMS Horizontal Gauge Deviation, Machine Learning Segmentation, Railroad Defect Detections, SAMURAI, YOLO

I. INTRODUCTION

During the past 200 years, railroads have played an important role in human society by enabling trade, providing affordable transportation, easing road congestion, reducing carbon emissions, and driving economic growth in the industrial and tourism sectors [1]. Ensuring the integrity and safety of a railroad system relies on timely detection of rail defects. One of the most critical parameters to monitor is the horizontal gauge between two tracks, including deviations such as gauge narrowing or widening. Horizontal gauge defects often de-

velop gradually due to factors such as sleeper degradation, clip failure, and rail wear. Because these changes occur slowly and incrementally, they are difficult to detect early with conventional inspection methods [2]. In light of significant advances in machine learning, computer vision, and image processing, more advanced methods have been introduced to reduce human labor in this task.

Prasetyo et al. [3] developed a manual mechanical gauge that uses Linear Motion Position (LMP) sensors attached to roller arms to measure the horizontal distance between two

railroad tracks as the device is pushed along the tracks. The method provides millimeter-level precision and consistent results, but is labor-intensive and requires precise alignment of the rollers with the rail heads during manual operation. In another paper by Moskal and Pastucha [4], traditional railroad track geometry measurements are described as contact-based systems using displacement sensors that physically follow the rails, providing accurate but slow and labor-intensive measurements that are unsuitable for continuous high-speed inspection. In contrast, the authors present a modern inertial-optical method that integrates Inertial Measurement Units (IMUs) with laser and camera sensors to enable non-contact, automated, and high-speed measurements, though it requires careful calibration and is sensitive to vibration and lighting conditions. The Rolling Track Gauge (Part No. RRT-4022-14) [5] manufactured by Railroad Tools and Solutions, Inc. is a manual measuring device designed to continuously monitor railroad track gauge as it rolls along both rails. The tool requires the operator to manually pull it along the track during inspection, allowing real-time detection of gauge variations through switches and crossings without interruption. Overall, these non-machine learning approaches demonstrate that achieving accurate railroad track geometry measurements often relies on labor-intensive manual operations.

Peng et al. [6] proposed a method for evaluating precise alignment of the track by combining machine vision and inertial data with an extended Kalman filter. Their multi-sensor system, using gyroscopes, accelerometers, inclinometers, and cameras, achieved sub-0.5 mm maximum deviations and under 0.3 mm standard deviations in real test conditions, demonstrating high precision, long distance measurement capability. Sresakoolchai and Kaewunruen [7] proposed a method that uses track geometry data collected by a Track Geometry Car (TGC) to detect rail defects along with other issues related to track components. They applied machine learning techniques including deep neural networks (DNN), convolutional neural networks (CNN), random forest (RF), support vector machines (SVM), decision trees (DT), multiple regression (MR), and gradient boosting (GB). Their best models achieved over 90% accuracy in identifying rail defects based on track geometry measurements. Ahmed [8] proposed a machine learning-based approach to predict track geometry using historical inspection data and time-series models. Their method applied functional networks and Long Short-Term Memory architectures to forecast parameters such as profile, alignment, and gauge, demonstrating accurate multi-step predictions over 6-12 months and enabling proactive maintenance planning in a 10-mile corridor.

Popov et al. [9] developed a pipeline that combines an autoencoder for anomaly detection and K-Means clustering to classify localized defects using over 15 years of high-speed track geometry measurements. By applying a moving frame of reference, their method accurately identifies regions of increased settlement, supporting targeted maintenance that reduces over-tamping and extends ballast lifespan. Pires et al. [10] proposed training a machine learning model to learn

a non-linear mapping between measured data, from an instrumented railroad vehicle (IRV), and track geometry to detect geometric irregularities. They simulated BRA1 vehicle dynamics to create training data, applied feature engineering and selection, and showed that while lateral irregularities were hard to predict, vertical irregularities could be estimated with root mean squared error (RMSE) reduced to 0.410 mm after post-processing. Jeripothula et al. [11] developed data-driven models for detecting railroad track misalignment that could lead to derailments, using a novel Track Misalignment Detection (TMD) dataset comprising images of normal and misaligned tracks. They framed the task as a binary image classification problem and applied transfer learning techniques. They also proposed an evaluation criterion to pre-select effective models, which their experiments showed outperformed other candidates during testing.

Tang et al. [12] conducted a systematic literature review of AI applications in railroad transport, analyzing 139 papers from 2010 to 2020 across sub-domains such as maintenance, planning, safety, and autonomous control. They found that research has focused mainly on maintenance and inspection, highlighting opportunities for future AI development in railroads. Kumar and Harsha [13] presented a comprehensive review of machine vision-based inspection systems (MVIS) as a solution to the inefficiencies of conventional railroad defect detection methods. They evaluated recent advances in image processing, computer vision, and deep learning to identify defects in railroad rolling stock and track infrastructure.

Although existing frameworks offer high accuracy, they typically require rich and detailed data from multiple expensive sensors to operate effectively, making full system deployment costly. Current railroad track gauge inspection systems are therefore limited by their dependence on multiple sensing components and high-cost infrastructure, restricting their accessibility and widespread adoption across different railroad operators. This study addresses this limitation by presenting a cost-effective approach to detect the horizontal track gauge using only data captured from an Intel D435 Depth Camera [14] mounted on a moving vehicle. Machine learning models are trained and deployed to process the collected data. The key contributions of this paper are:

- 1) Identifying the most suitable machine learning model with optimized modifications to detect running tracks. A custom-labeled dataset with 3,819 augmented images is created to train and evaluate segmentation models. The dataset includes images collected from both public railroads and a security-sensitive military base. Due to access restrictions, only the publicly collected portion, consisting of 1,164 images prior to augmentation, is available for sharing¹.
- 2) Developing an accurate technique to measure the distance between tracks using depth-based 3D reconstruction, including a denoising pipeline to handle depth noise from vegetation and motion.
- 3) Modifying the SAMURAI segmentation model to support multiple-object tracking using multiprocessing and

memory optimization. The source code for the SAMURAI modification, data extraction, denoising pipeline, and distance measurement is publicly available in a GitHub repository².

- 4) Analyzing and comparing the tracking performance of different segmentation models: YOLOv5x-seg, YOLOv8x-seg, YOLOv9e-seg, YOLO11x-seg, and SAMURAI.

Our results demonstrate that deep learning models can reliably detect horizontal gauge from a moving platform, offering a promising low-cost alternative for railroad defects detection. This system distinguishes itself through the integration of several practical engineering improvements. These include a denoising algorithm developed to reduce measurement errors caused by vegetation and motion, a customized version of the SAMURAI tracker capable of processing both rails simultaneously, and a 3D measurement approach that pairs track edges to enable accurate gauge calculation. This integrated approach provides a practical and cost-effective alternative to traditional, sensor-intensive inspection techniques.

The remainder of this paper is organized as follows. Section II discusses machine learning models. Section III outlines the image processing techniques and the measurement method. Section IV explains the limitations of the current measurement method and reviews different attempts to address them. Finally, Section V presents the conclusions and directions for future work.

II. MACHINE LEARNING MODELS

Among the many available segmentation models, YOLO-seg, a segmentation-focused extension of the original You Only Look Once (YOLO) [15] object detection model, and SAMURAI [16], a modified version of Meta's SAM2 [17] segmentation model that incorporates Motion-Aware Memory for improved object tracking, were selected for evaluation. YOLO-seg was chosen for its real-time performance and high detection accuracy, while SAMURAI was selected for its robust tracking capability and zero-shot segmentation ability.

A. YOLO-SEG

YOLO is a deep learning framework for object detection that predicts bounding boxes and class probabilities in a single pass of an image. Earlier YOLO models focused only on object detection, without segmentation capability. With YOLOv5 [18], Ultralytics introduced integrated segmentation support, allowing the model to produce pixel-level masks alongside the bounding boxes. This capability is particularly useful for railroad scenes, where precise separation of rails from ballast, sleepers, or surrounding vegetation is essential for accurate horizontal gauge measurement. YOLOv8 [19] builds on this foundation with an improved CSPDarknet backbone for richer feature extraction and an anchor-free, decoupled detection head that independently

handles classification and localization. These changes can help the model better distinguish long, narrow rail structures and reduce confusion from background textures, such as gravel or shadows. YOLOv9, developed by Wang et al. (Academia Sinica) [20], introduces Programmable Gradient Information (PGI) and the Generalized Efficient Layer Aggregation Network (GELAN) to preserve fine-grained spatial information and improve feature flow. These mechanisms may improve segmentation consistency and rail edge clarity, leading to more stable segmentation results across frames under varying lighting conditions. YOLOv11 [21], the most recent Ultralytics release at the time of writing, further enhances detection speed and precision with an optimized backbone and neck, a low-latency detection head, and seamless deployment across edge devices, cloud platforms, and NVIDIA GPUs. These updates help reduce inference time and simplify model integration, making implementation more practical for large-scale or automated processing workflows. These ongoing improvements have enabled diverse real-world applications that capitalize on the real-time and high-accuracy characteristics of YOLO. For example, YOLOv8-seg has been used to improve precision and efficiency in the laser cleaning of Printed Circuit Boards (PCBs) [22], and a lightweight YOLOv8-seg variant has been proposed for real-time pixel-level crack detection essential to structural health monitoring [23]. In this paper, we evaluated Ultralytics' YOLO versions (YOLOv5, YOLOv8, and YOLOv11) and YOLOv9, which is supported but not originally developed by Ultralytics, due to their straightforward training and strong community support.

B. SAMURAI

SAMURAI is an advanced segmentation model derived from SAM2, an evolution of Meta's Segment Anything Model (SAM) [24]. The original SAM pioneered promptable zero-shot segmentation for static images. It allows users to provide simple prompts, such as points or boxes, and uses the provided prompt to segment any object without additional training. SAM2 extended this idea to video by introducing a streaming memory mechanism that maintains context across frames for real-time segmentation. To further improve segmentation and tracking performance in harsh or dynamic environments, SAMURAI introduces Motion-Aware Memory, which intelligently selects and retains the most relevant past frames based on mask quality and motion cues. Moreover, it incorporates a Kalman filter-based motion model to better predict object trajectories over time. This helps maintain consistent object localization even in challenging conditions. Specifically, SAMURAI chooses the most promising mask for each frame by combining the segmentation affinity score with the Kalman filter-predicted IoU score in a weighted manner, improving robustness in difficult tracking scenarios. Figure 1 illustrates an example of selecting the most suitable mask with the highest final score. These enhancements make SAMURAI particularly well-suited for applications where maintaining object identity across frames is critical.

¹universe.roboflow.com/erdc-rail-line-dataset/railroad-ballast-qgzvt

²github.com/chybu/horizontal-gauge-measurement

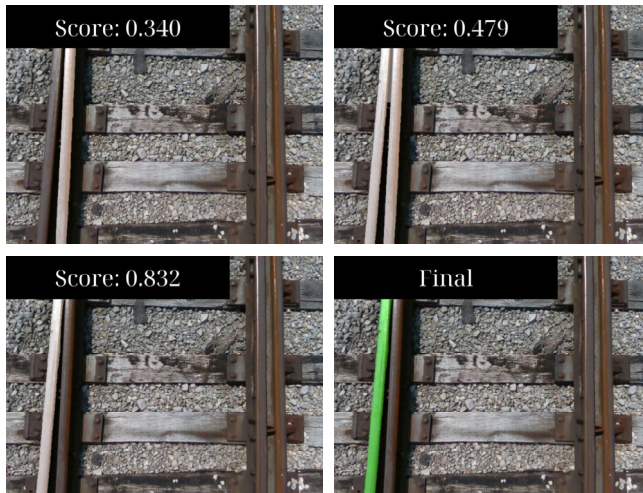


FIGURE 1. Candidate Masks by SAMURAI with their Scores

III. METHODOLOGY

A. MODEL EVALUATION AND SELECTION

1) YOLO Versions

Four YOLO-based segmentation models, YOLOv5x-seg, YOLOv8x-seg, YOLOv9e-seg, and YOLO11x-seg, were trained for 100 epochs each on an NVIDIA A100 GPU. The training was conducted using a dataset that initially comprised 1,552 annotated images labeled with two classes: 'railroad' and 'ballast'. The inclusion of the ballast class follows the recommendation of Zhou et al. [25]. These researchers emphasized that by labeling additional classes, including background and secondary categories, they provide contextual information that enhances the discrimination of the primary target class. Among the initial dataset, 346 images contain vegetation that either lies adjacent to or partially overlaps with the railroad tracks. These images help the models learn under natural occlusion and clutter conditions, improving their robustness in field environments. Furthermore, data augmentation was applied to improve model robustness, including exposure variation ($\pm 3\%$), Gaussian blur (up to 2.5 pixels), and salt-and-pepper noise (up to 0.22% of pixels). The final dataset consists of 3,819 images, split into 89% for training (3,400 images), 7% for validation (278 images), and 4% for testing (141 images).

TABLE 1. Performance Comparison of YOLO Segmentation Models for Railroad Class

Version	Precision	Recall	mAP@50	mAP@50:95	F1-score
YOLOv5x-seg	0.968	0.960	0.975	0.782	0.964
YOLOv8x-seg	0.931	0.959	0.965	0.803	0.945
YOLOv9e-seg	0.899	0.973	0.971	0.799	0.935
YOLO11x-seg	0.913	0.960	0.963	0.785	0.936

Table 1 shows the validation results of the trained YOLO models, focusing only on the 'railroad' class. YOLOv5x-seg achieves the highest precision (0.968), mAP@50 (0.975), and F1-score (0.964), indicating strong overall segmentation per-

formance despite limited data. YOLOv8x-seg has the highest mAP@50:95 (0.803), suggesting improved mask alignment at stricter IoU thresholds, but with slightly reduced precision and F1-score that may reflect sensitivity to small-scale data variability. YOLOv9e-seg achieves the highest recall (0.973), demonstrating effective detection of all relevant regions but at the expense of increased false positives, leading to lower precision. YOLO11x-seg yields consistently lower metrics, likely due to the greater complexity of its architecture, which may require larger and more diverse datasets to train effectively. Given these results, YOLOv5x-seg was selected for deployment due to its superior overall performance, simplicity, and robustness when applied to a limited domain-specific segmentation task.

When tested on a dataset of unused railroad tracks with significant vegetation, the trained YOLOv5x-seg model can successfully detect the tracks, as shown in Figure 2. However, it also has the potential to produce incorrect masks, as illustrated in Figure 3. This limitation suggests that, although the training dataset includes vegetation-rich scenes, the number of such samples may still be insufficient for the model to fully generalize under dense vegetation conditions, leading to segmentation errors where track regions are confused with background areas such as ballast or vegetation. A notable issue, which is closely related to YOLO's inherent strength in exhaustive detection, is its tendency to identify every visible track segment, resulting in overlapping masks and irrelevant segmentations (Figure 4). This behavior complicates the task of reliably selecting the correct pair of tracks for measurement. Consequently, instead of serving as the primary model for track detection, YOLOv5x-seg will instead be used as an auxiliary tool to support SAMURAI.



FIGURE 2. Examples of Original Images and Corresponding Segmentation Masks Generated by YOLO

2) SAMURAI Modification

Unlike the YOLO model, which automatically detects trained object classes without additional input, SAM2, the core

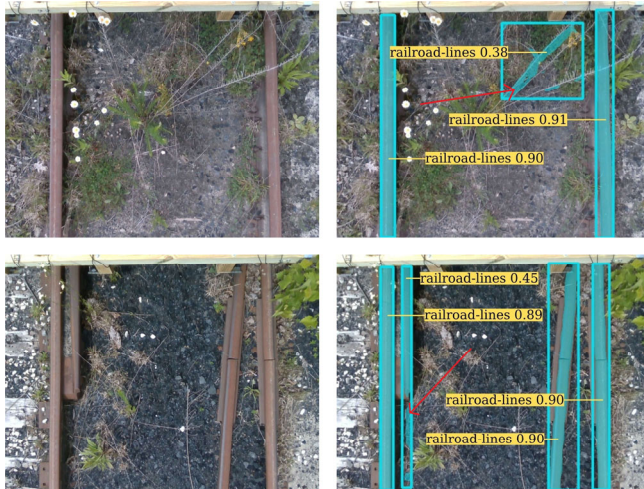


FIGURE 3. Examples of Original Images and Corresponding Incorrect Segmentation Masks Generated by YOLO



FIGURE 4. Examples of Original Images with Overlapping Masks and Irrelevant Segmentations Generated by YOLO

model underlying SAMURAI, requires an initial prompt to generate the segmentation mask. Although YOLO's tracking performance is relatively poor, its output can serve as an effective initialization for SAM2 or SAMURAI. SAMURAI exhibits superior tracking performance compared to YOLO as shown in Figure 5. However, this comes at the cost of increased memory consumption. In the original paper [16], SAMURAI is described as using a selective motion-aware memory mechanism that retains only the most informative frames rather than storing all previous frames. However, in the published code implementation, although tracking computations may be based on a limited frame range, the system still stores the tracking results of all frames, including predicted masks, Kalman filter scores, IoU scores, and other metadata, without clearing unused data [26]. This accumulation can become unnecessarily large and lead to out-of-memory er-

rors during long video sequences. To address this issue, the SAMURAI model was modified to retain only the most recent 100 frames, thereby maintaining a fixed memory footprint regardless of sequence length.

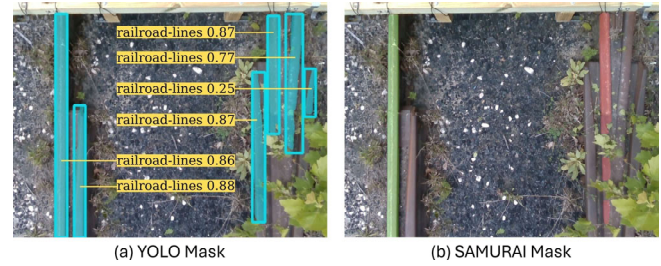


FIGURE 5. Comparison of YOLO Mask and SAMURAI Mask Segmentation Results

Since the current SAMURAI model supports tracking only one object at a time, it cannot natively track both railroad tracks simultaneously, an essential requirement for measuring the distance between them. To address this limitation, multiprocessing is implemented with two separate processes, each responsible for tracking one railroad track and storing its results in shared memory. Running on an NVIDIA T4 GPU, the two processes together utilize approximately 12.7 GB of GPU memory, with an average processing time of around 0.85 seconds per frame.

A further limitation of SAMURAI arises from its reliance on prior frames for tracking. It cannot seamlessly switch to a new object when the current target disappears (see Figure 6), even though it may eventually detect the new object after a few additional frames. In Figure 6, the right railroad track disappears over frames as it merges with another track. Even when the old track nearly disappears, it still does not switch to the new one, and eventually SAMURAI loses track until several additional frames later. In such cases, SAMURAI fails to return any masks until the new object is detected. This situation can occur, for example, when two railroad tracks merge into one, or when a single track splits into two. To address this issue, SAMURAI's re-prompting function can be used, with the YOLO model providing new prompts for emerging objects. Although not implemented in this study, this feature helps maintain consistent detection masks as tracked targets change over time and will be considered for integration in future work.

B. DATA PROCESSING

The data used in this study were captured using an Intel D435 Depth Camera mounted on a moving vehicle. The camera's native output format is a .bag file, which stores color frames, depth frames, camera settings, and calibration data, as well as timestamps for all frames. This results in extremely large file sizes even for short recordings (e.g., 14.8 GB for an 8-minute video). To reduce storage requirements and processing overhead, rather than using the full .bag files directly, frames are extracted at specified intervals. This interval can be adjusted based on the vehicle's speed: faster

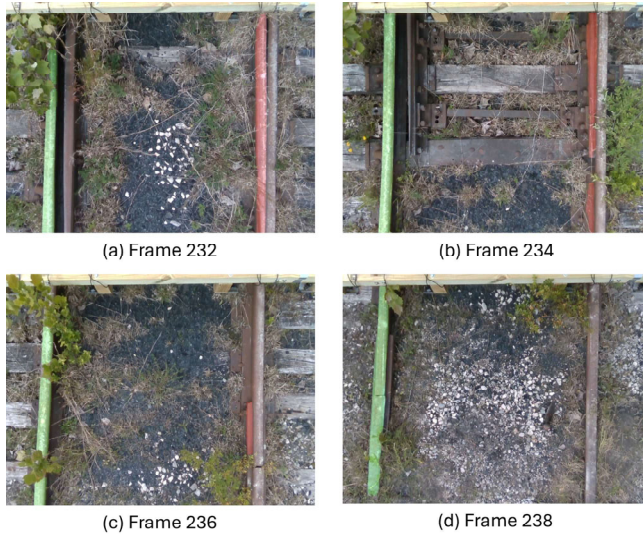


FIGURE 6. SAMURAI Not Switching from the Right-Side Railroad Track to the Emerging Track (Frames 232-238)

movement requires smaller intervals to avoid missing sections of the track. For the dataset used in this paper, captured at approximately 5 miles per hour, an interval of 6 frames was selected, meaning information was extracted every 6 frames. Three types of data are extracted: color frames, stored as JPEG images; depth frames, saved as 16-bit grayscale PNG images with each pixel representing depth in millimeters; and camera settings, recorded in text files. This extraction strategy reduces the original dataset size from 14.8 GB to approximately 600 MB, representing a substantial reduction while preserving the necessary information for analysis.

The recorded depth data are inherently noisy due to factors such as vegetation along the railroad, sunlight interference, and vehicle-induced motion, necessitating effective denoising prior to accurate distance measurement. To address this, a median blur filter with a kernel size of 1×5 pixels is first applied to remove isolated noisy outlier pixels while preserving the distinction between the elevated railroad surface and the ground. Missing depth values (encoded as zeros) are then identified and replaced with the average of neighboring depth values to fill gaps in the data. A significant challenge stems from vegetation, which can severely disrupt measurement by introducing anomalous depth readings. To mitigate this, a strong Gaussian blur with a kernel size of 1×401 pixels (favoring the vertical direction) is applied to produce a baseline version of the depth image in which the vegetation is completely smoothed out. The difference between this baseline and the original depth image is then computed, and any pixel whose difference exceeds a predefined threshold is corrected by substituting the baseline depth value. Finally, three additional Gaussian blurs with progressively increasing kernel sizes of 1×31 , 1×51 , and 1×101 pixels are applied sequentially to achieve controlled, gradual smoothing of the denoised depth map. Figure 7 provides an example of the

depth data before and after the denoising process, where the blue regions represent pixels that were modified during the denoising process. Figure 8 illustrates the difference between the measurements obtained with and without denoising. Appendix A shows the mathematical model of the denoising pipeline.

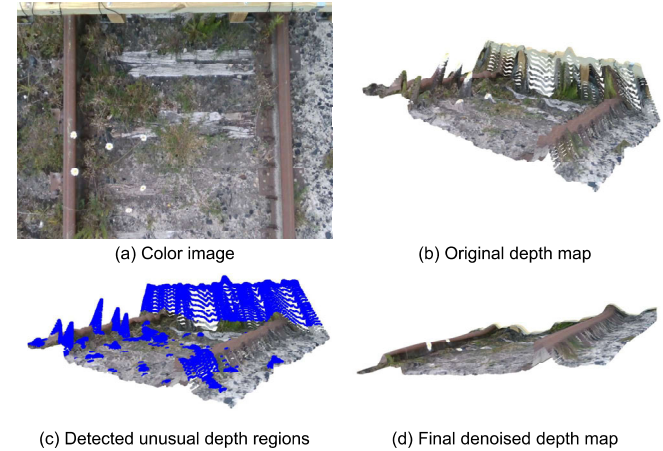


FIGURE 7. Illustration of the Denoising Strategy

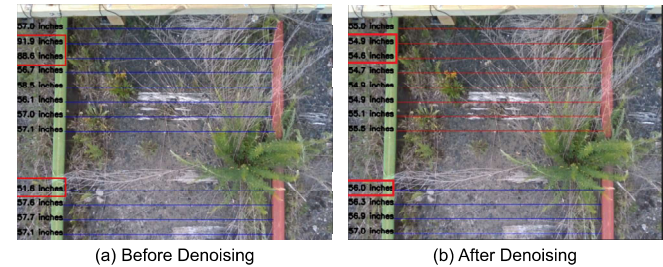


FIGURE 8. Difference in Measurement Results, Measured in Inches, Without and With Denoising

C. MEASUREMENT METHOD

By combining the color image, depth image, and the camera intrinsics file, which contains intrinsic parameters such as ppx, ppy, fx, and fy, the scene can be projected into three-dimensional space. Incorporating the segmentation results from SAMURAI enables accurate localization of the railroad tracks within this 3D representation. With this information, the distance between the two railroad tracks can be measured. As stated in an official document published by the United States Department of Defense (2018), the standard gauge in the United States is 56.5 inches [27]. This measurement refers to the distance between the inner faces of the two railroad tracks. For the left railroad track, the inner face corresponds to the points along the left edge of the segmentation mask, while for the right railroad track, it corresponds to the points along the right edge of the mask. Then, the two masks will crop out extra parts so that they can have the same size (illustrated in Fig. 9). This process produces two separate sets of points, where each point in one set connects to a point having the

(a) 3D representation

(b) 2D projection

```
graph TD
    Start([Start]) --> ExtractData[Extract data from the original .bag file with a frame interval]
    ExtractData --> ExtractColor[Extract the color image as .jpg file]
    ExtractData --> ExtractDepth[Extract the depth image as 16-bit grayscale PNG]
    ExtractData --> ExtractSettings[Extract camera settings as .txt file]
    ExtractColor --> YOLO[Use YOLO model to create initial prompt for SAMURAI]
    YOLO --> SAMURAI[Use SAMURAI to track the railroads]
    SAMURAI --> Pairs[Extract pairs of points from the segmentation masks]
    ExtractDepth --> Denoise[Denoise depth data with hole filling, median blur, and Gaussian blur]
    Pairs --> Transform[Transform 2D points to 3D points]
    Denoise --> Transform
    Transform --> Distance[Calculate the Euclidian distance between the two points in each pair]
    Distance --> Decision{Is the calculated distance out of safety range?}
    Decision -- YES --> Output[Horizontal Gauge Deviation detected  
Report GPS location for manual inspection.]
    Decision -- NO --> End([End])
```



Figure 13 demonstrates the measurement results across four consecutive frames with a 6-frame interval. The recorded distances are overall consistent. Manual measurements from different sections of the railroad, as shown in Table 2, were used to establish a reference baseline. The manual measurements yielded a mean gauge of 56.6 inches with a standard deviation of 0.27 inches, indicating minimal variation and confirming the uniformity of the reference track section. This mean value was therefore adopted as the baseline for calculating the percentage error. The average measured distance in Figure 13 is 57.3 inches. Consequently, the percentage error

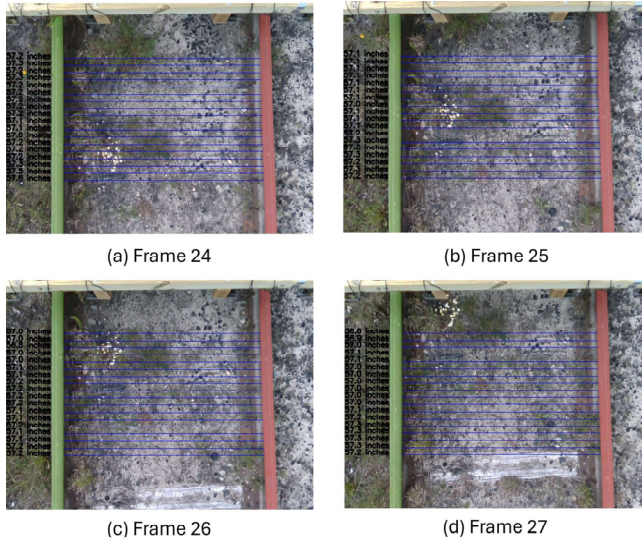


FIGURE 13. Measurement Results over Four Consecutive Frames with a 6-frame Interval, Measured in Inches

is

$$\frac{|57.3 - 56.6|}{56.6} \times 100 = 1.2\%.$$

This percentage error is higher than the expected value of 0.56%, which was obtained under controlled laboratory conditions with a stationary vehicle. The elevated error of field data can be attributed to the challenges inherent in outdoor measurement from a moving vehicle, such as vibrations and environmental variability.

TABLE 2. Field-measured Track Gauge Values in Inches

Gauge (inches)
56.80
56.60
56.30
56.25
57.00
57.00
56.50
56.60
Mean = 56.63
Standard Deviation = 0.27

Measurements were also recorded in which the algorithm detected horizontal gauge deviation, as shown in Figure 14. Figure 14 illustrates the measured distance between two tracks in inches. If the measured distance falls within the safety range (55.69-58.07 inches), the distance line is shown in blue. Otherwise, it is shown in red. However, the detections shown in this figure are false positives that arise from two main reasons. First, the generated segmentation mask excludes vegetation on the railroad. Second, the original depth values are heavily distorted by the presence of vegetation. As a result, the measured distance deviates from the actual distance. A detailed analysis of the vegetation's impact on the

measurement will be discussed in the Results and Limitations section. To avoid distortion caused by vegetation and better evaluate the algorithm's performance, synthetic railroad tracks with intentional horizontal gauge deviations were constructed in a controlled laboratory environment. The complete measurement pipeline, including denoising and multi-process tracking, was applied. The safety range for the laboratory setup was defined as 45 to 47 inches, and measurements falling outside this range were classified as exhibiting horizontal gauge deviation. Figure 15 demonstrates that the algorithm effectively detected deviations in cases of altered track gauge, while correctly recognizing normal conditions when the gauge was within the defined safety range. These results demonstrate that the fully integrated algorithm performs as intended in a controlled laboratory environment.

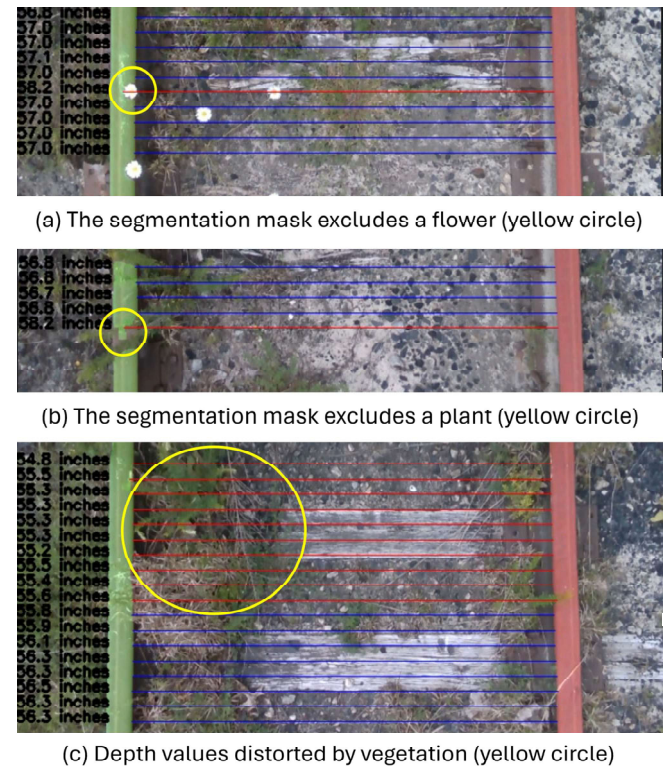


FIGURE 14. Example of Detected Horizontal Gauge Deviation, Measured in Inches, Caused by Vegetation Effects (Red Lines)

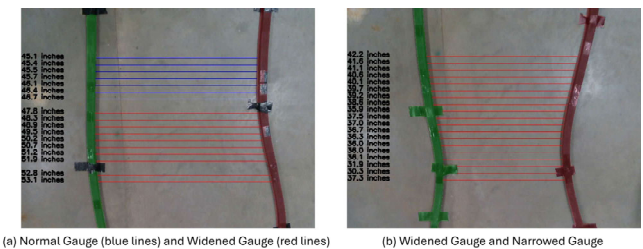


FIGURE 15. Results of the Proposed Algorithm Showing Detection of Horizontal Gauge Deviations (Red Lines) on Synthetic Railroad Tracks Under Laboratory Testing Conditions

2) Machine Learning Model Tracking Performance

Figure 16 and Figure 17 show the tracking performance of the YOLOv5x-seg and SAMURAI segmentation models over video frames, measured by two key metrics: (a) the number of predicted masks per frame and (b) the average confidence score per frame. Ideally, the system should maintain a consistent prediction of two rail masks per frame (representing the left and right tracks) with high confidence scores near 1.0.

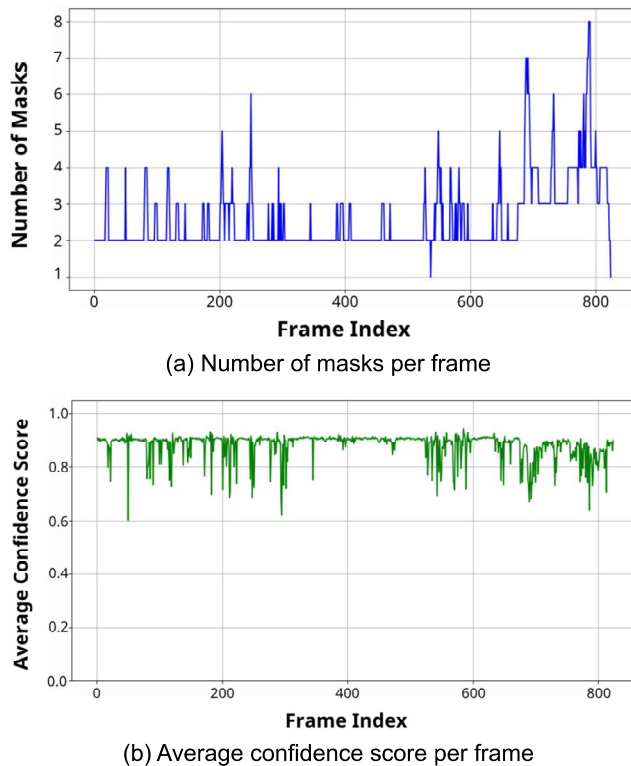


FIGURE 16. YOLOv5x-Seg Segmentation Variability and Confidence Fluctuations Across Video Frames Under Visually Complex Conditions

As shown in Figure 16, YOLOv5x-seg exhibits substantial variability in the number of masks per frame, frequently exceeding two and occasionally spiking as high as eight. This behavior reflects its tendency to produce fragmented and overlapping segmentations in complex scenes with vegetation, shadows, or track switches, as shown in Figure 3 and Figure 4. While its average confidence scores are 0.84, there are notable dips below 0.8 in some frames, indicating reduced segmentation certainty in cluttered or occluded regions. These results highlight YOLO's sensitivity to scene complexity and its potential to introduce extra false-positive segments, which can complicate reliable gauge measurement.

In contrast, Figure 17 demonstrates that the SAMURAI model achieves markedly more stable tracking. The number of predicted masks per frame remains consistently close to the ideal value of two across most of the video sequence, even when average confidence occasionally dips below 0.8 due to vegetation or track switches. This suggests that SAMURAI's motion-aware memory allows it to maintain consistent left-right rail tracking despite transient challenges like occlu-

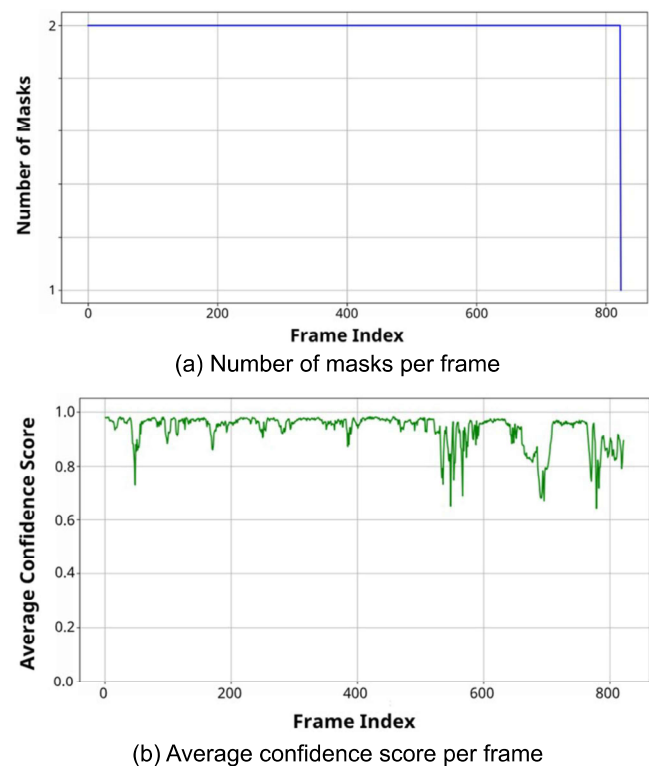


FIGURE 17. SAMURAI Segmentation Stability and Confidence Consistency Across Video Frames Under Visually Complex Conditions

sion or motion blur. The primary cause of the fluctuation in SAMURAI's and YOLO's confidence scores is the presence of vegetation. As shown in the average confidence score per frame plot for both models (Figure 17 and Figure 16), there are specific frame ranges where both models exhibit a noticeable drop in confidence. This decreasing range, highlighted in Figure 18, corresponds to the sections of the video in which vegetation appears in the scene, affecting the models' ability to reliably segment or detect railroad tracks.

3) Impact of Vegetation on False Positive Detection

A test was conducted to evaluate the effect of vegetation on false positive detection. Figure 18 shows that frames containing vegetation have reduced confidence scores. Based on this observation, two representative frame ranges were selected for quantitative analysis. The first range (frames 400-500) corresponds to a section with minimal vegetation, while the second range (frames 550-650) represents a section with severe vegetation. The statistical results of the two test ranges are summarized in Table 3.

The non-severe vegetation range exhibits consistent measurements near the standard gauge value, with small standard deviation indicating stable model performance under clear conditions. In contrast, the severe vegetation range shows a 13 times higher standard deviation and a mean value closer to the lower bound of the safety range, reflecting greater measurement noise and a tendency toward underestimation. These findings confirm that dense vegetation often causes

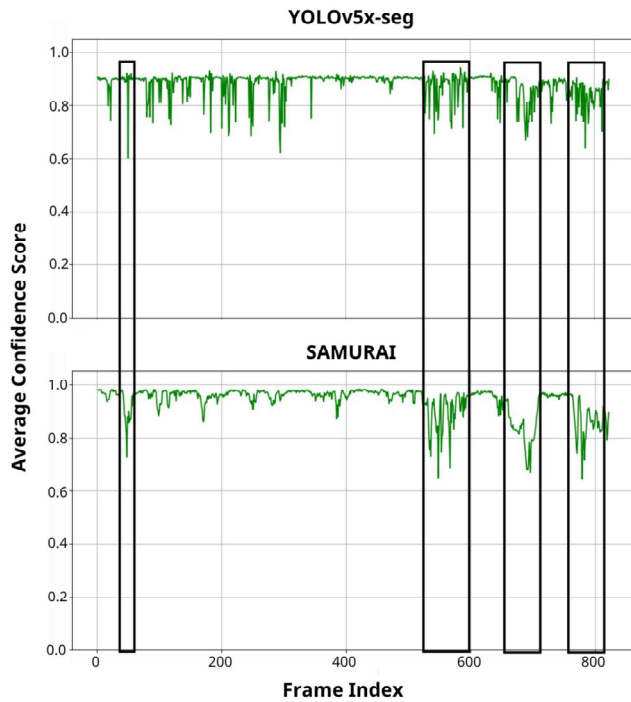


FIGURE 18. Frame Ranges with Decreased Confidence Scores in SAMURAI and YOLO Due to Vegetation

TABLE 3. Vegetation Test Results for Different Track Conditions

Metric	Non-severe Vegetation	Severe Vegetation
Frame range	400–500	550–650
Number of frames	100	100
Detected frames	5	10
Mean (in)	57.49	56.41
Median (in)	57.57	57.06
Standard deviation (in)	0.43	5.78

underestimation of the measured rail distance due to occlusion and visual interference in both color and depth data. This underestimation ultimately leads to false positive detections of gauge deviation.

B. LIMITATIONS

Overall, the result analysis demonstrates that combining YOLOv5x-seg for initial detection with SAMURAI for temporal tracking provides stable and accurate segmentation under diverse conditions. Despite these advantages, several limitations remain in the current system, as discussed below.

1) Vegetation Interference and Measurement Accuracy

As mentioned above, this measurement method can produce false positives for two main reasons. First, the generated segmentation mask may exclude vegetation on the railroad. Secondly, the original depth values can be greatly distorted by the presence of vegetation. Although the denoising pipeline reduces some of this interference, it is not always sufficient to prevent gauge inaccuracies. To mitigate this limitation, several deep learning models, including YOLOv8-seg [19],

U-Net [28], and DeepLabv3+ [29], were trained to detect vegetation in the measurement area. The underlying idea is that when vegetation is detected within the measurement area, the corresponding measured distance is visually highlighted with a green line, in contrast to the red line used to indicate horizontal deviation, thereby enhancing user awareness. In addition to providing a warning when vegetation is present, detected vegetation regions can also be excluded from the measurement process if they overlap with the railroad area, thereby reducing the influence of occlusion and improving gauge accuracy. Among these models, DeepLabv3+ produced the most accurate and reliable segmentation results. An example of an original image and its corresponding segmentation mask generated by DeepLabv3+ is shown in Figure 19. However, because of its complex architecture and relatively long inference time, integrating it would noticeably increase the overall processing duration. Future work will focus on implementing a lightweight vegetation detection model to preserve warning capability while maintaining real-time processing efficiency. In addition, data augmentation techniques such as lighting variation, exposure adjustment, rotation, flipping, and noise addition will be employed to simulate diverse vegetation scenarios and improve model robustness. Vegetation detection latency will also be evaluated in detail to identify and explore optimization strategies for enhanced overall performance.

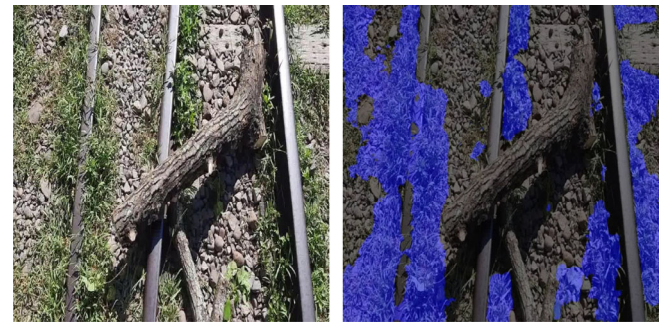


FIGURE 19. Examples of Original Image and Corresponding Segmentation Mask Generated by DeepLabv3+

2) Dataset Diversity and Environmental Limitations

The dataset used in this study comprises a diverse collection of images captured under varying lighting conditions; mainly recorded during daytime hours and across different trackbed materials (see Figures 1, 2, and 3). However, it does not include any curved track segments. Despite this limitation, the proposed method remains robust and performs reliably even in such scenarios. This is because each frame captures only a small portion of the railroad, which makes the rails appear locally straight within the measurement area even when the track itself is curved. The effect is similar to observing a small section of a larger curve, where the local geometry remains nearly linear. The extent of the captured area depends on the height of the camera. In this study, the camera was placed

approximately 11.5 ft (3.5 m) above the ground, capturing an area of about 100 inches in width and 71 inches in height. Within this field of view, the curvature of a typical railroad is minimal relative to the measurement precision, ensuring that gauge estimation remains accurate even when the rail lies on a curved section.

Although the dataset includes scenes with varying levels of vegetation, this variation is not sufficient to ensure a robust generalization, as the trained YOLO model still struggles to detect the railroad in certain cases, as shown in Figure 3. To enhance robustness and reduce potential overfitting, future data collection may include images captured under distinct environmental characteristics. Without such improvements, the current YOLO model used for initial detection may not perform well in varied environments.

3) Sensitivity to Camera Orientation

Another important limitation is that the method assumes a specific operating condition in which the camera orientation is aligned with the moving direction. In general, two types of angular deviations can occur: the pitch angle, which represents the vertical tilt of the camera (upward or downward), and the side rotation angle, which represents the horizontal deviation between the camera orientation and the actual moving direction. While changes in pitch angle have minimal influence on detection accuracy, variations in the side rotation angle can introduce measurable errors in the estimated rail distances. To evaluate sensitivity to this side misalignment, the full measurement pipeline was applied to three video sequences corresponding to different rotation angles, as illustrated in Figure 20. From left to right, Figure 20 shows three moving direction scenarios of the same testing environment. In the first case, the camera orientation and moving direction are aligned, corresponding to a rotation angle of 0° . In the second case, the moving direction is angled 31° to the right of the camera orientation. In the third case, the moving direction is angled 8° to the left of the camera orientation. Testing results are shown in Table 4.

TABLE 4. Effect of Camera Orientation Misalignment on Measured Gauge

Test Case	Average Gauge (in)	Error Rate (%)
0° (Aligned)	35.99	0.0
8° (Left)	36.51	1.4
31° (Right)	42.62	18.4

The error rate E_β for a rotation angle β was computed as

$$E_\beta = \frac{|G_\beta - G_0|}{G_0} \times 100\%,$$

where $G_0 = 35.99$ inches is the baseline gauge at 0° and G_β is the average gauge at angle β . Assuming a quadratic dependence between error and rotation angle, i.e.,

$$E_\beta = k \beta^2,$$

the proportionality constant k can be obtained using either measurement pair. Using the 31° case, $k = \frac{18.4}{31^2} = 0.0192$,

and using the 8° case, $k = \frac{1.4}{8^2} = 0.0219$. Averaging the two gives $k \approx 0.0205$. Therefore, the empirical relationship can be expressed as

$$E_\beta \approx 0.0205 \beta^2\% \quad (\beta \text{ in degrees}),$$

which predicts about 1.3% at 8° and 18.7% at 31° , consistent with the measured 1.4% and 18.4%. This relationship shows that even moderate deviations between the camera orientation and moving direction introduce significant geometric distortion in the measured rail distance. To address this problem, several methods have been tested, among which the Rotate Angle technique (described in the following Rotate Angle section) effectively mitigates this error under standard operating conditions, as illustrated in Figure 23. This technique also enables detection of gauge deviation, even when the distance measurement is partially inaccurate, in more complex scenarios such as the one shown in Figure 24.

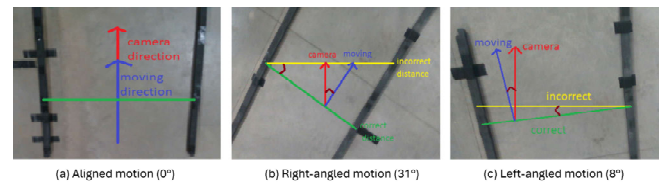


FIGURE 20. Illustration of Measurement Error Due to Camera Misalignment with Motion Direction

a: Closest Distance Between Two Points

Instead of identifying pairs with the same y-value, an alternative approach is to find pairs of points that have the smallest Euclidean distance in a 3D space. However, this method introduces a significant challenge, as many different points can end up selecting the same single point as their closest neighbor. This results in a "triangle pattern," as illustrated in Figure 21, where the points on the right track serve as the source point set and the points on the left track serve as the destination point set. Consequently, this approach does not provide a reliable or consistent solution for accurate gauge measurement.

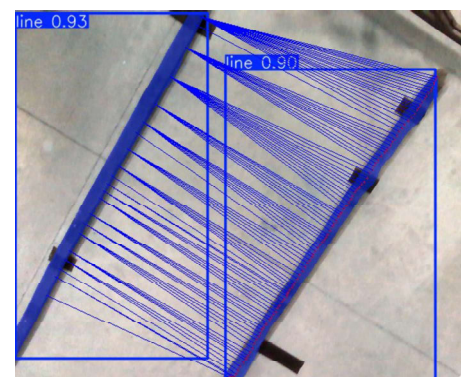


FIGURE 21. Example of a Triangle Pattern Formed When Points on the Right Track are Paired to the Closest Points on the Left Track

b: Rotate Angle

In the ideal case where the camera's viewing direction is perfectly aligned with the vehicle's direction of motion, the two railroad tracks can be approximated as lines whose angles relative to the vertical axis are symmetric. However, in non-ideal scenarios, these angles become asymmetric due to camera misalignment. The rotation angle is computed to realign the image so that the track angles are symmetric with respect to the vertical axis. This correction ensures that points on the left and right rails share matching y-coordinates, enabling accurate pairing. Alternatively, instead of rotating the entire image, this rotation can be applied directly to the paired points. In the ideal case, these pairs have zero angle relative to the horizontal axis. In the angled case, the pairs exhibit a consistent angle equal to the required rotation correction, as illustrated in Figure 20.

First, skeletonization is applied to the segmentation mask to generate a set of representative points for each track. From these points, the best-fit line corresponding to each track is determined. Using the function of the best-fit line, the angle to the vertical axis can be calculated as $\theta = \arctan\left(\frac{1}{\text{slope}}\right)$. The rotation angle is then computed as the average of these two angles, $\text{rotation angle} = \frac{\theta_1 + \theta_2}{2}$. Finally, the image is corrected by applying the final rotation given by final rotated angle = original angle - rotation angle. This process is illustrated in Figure 22. An example result of this method is shown in Figure 23, which displays both the correctly rotated version of the original image and the rotation-corrected pair points. Although effective in simple cases, this method struggles in situations where the railroad cannot be accurately represented by a first-degree function. For example, in Figure 24, even in an ideal scenario with no camera misalignment relative to the direction of motion, the algorithm incorrectly detects an angled case and produces inaccurate point pairings. However, despite incorrect measurement of reference points, the algorithm still successfully detects the gauge distance deviation. Moreover, due to the temporal nature of video capture, this issue is mitigated in subsequent frames, where the railroad tracks become more clearly represented and can be accurately modeled using a first-degree function.

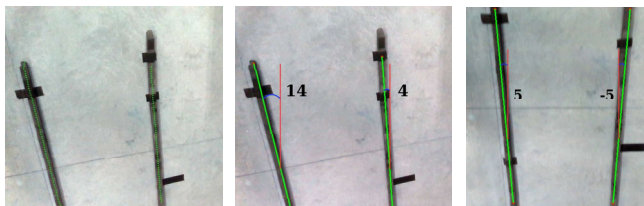


FIGURE 22. Illustration of the Rotation Correction Process for Aligning Track Masks

V. CONCLUSION

1) Summary

This study demonstrated a cost-effective approach to detecting horizontal gauge deviations in railroad tracks using an



FIGURE 23. Example of Rotation Correction Applied to the Original Image and to Paired Points

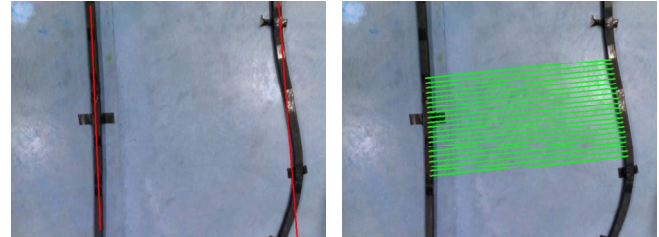


FIGURE 24. Example of Incorrect Point Pairing in an Ideal Scenario Without Camera Misalignment

Intel D435 Depth Camera mounted on a moving vehicle. Using YOLOv5x-seg and SAMURAI segmentation models combined with depth-based 3D reconstruction and a denoising pipeline, the proposed method enables measurement of rail distances with promising accuracy. Field measurements showed an average error of 1.2%, reflecting challenges posed by outdoor conditions such as vegetation, lighting variation, and vehicle motion. Limitations include false positives caused by vegetation-related underestimation of rail distances, the limited size and diversity of the training dataset, and the reliance on camera alignment with the motion direction. To address the camera angle problem, two techniques were developed: the Closest Distance Between Two Points method and the Rotate Angle method. The latter effectively mitigates measurement errors under standard operating conditions and enables detection of horizontal gauge deviations, even with partially inaccurate distance measurements, in more complex scenarios. In addition, a synthetic railroad environment was introduced to evaluate the measurement pipeline in controlled conditions, since heavy vegetation in real environments made field validation difficult. Although the results confirm the feasibility of low-cost, camera-based horizontal gauge detection, further improvements in segmentation robustness, vehicle dynamic calibration, and adaptability to varied track environments are needed to enable practical large-scale deployment.

2) Future Work

Future work will primarily focus on three aspects: camera angle correction, vegetation detection, and dataset expansion. First, to improve geometric accuracy under varying orientations, an Inertial Measurement Unit (IMU) rigidly mounted to the depth camera will be integrated to provide real-time orientation data, particularly yaw, for correcting horizontal misalignment caused by camera rotation relative to the track. By combining this input with depth-based 3D reconstruction,

the system can dynamically re-align measurements and maintain accuracy even when the vehicle or camera orientation drifts. Second, efforts will focus on developing a vegetation detection model and evaluating its detection latency to achieve an optimal balance between accuracy and real-time performance. The goal is to improve the system's capability to identify vegetation under diverse environmental conditions and reduce false positive alerts that affect gauge estimation. Third, future data collection will aim to expand and diversify the dataset by including images from multiple geographic regions, track materials, and illumination conditions, thereby enhancing the model's generalization and reliability in practical railroad applications.

Although this study focuses on horizontal gauge measurement, the same depth-based framework has the potential to support vertical track alignment analysis. The denoising pipeline currently identifies and suppresses vertical anomalies such as vegetation or debris on the track surface by detecting abrupt changes in depth values. With additional tuning, this capability can be repurposed to detect vertical misalignments, such as track dips, and uneven elevation. Such anomalies are important indicators of structural instability and can affect train safety and ride comfort. Future work will explore algorithms to quantify these height variations, distinguish between noise and true elevation defects, and incorporate them into a comprehensive track health monitoring system.

3) Broader Applicability

The core concept of this framework is to use RGB images for object segmentation and depth data for three-dimensional reconstruction to calculate the distance between segmented structures. This principle can be extended to other fields that require accurate measurements of the distance between components. The segmentation target can be adjusted depending on the application. If higher precision is required, the system can integrate a more advanced RGB-D sensor or a supporting measurement instrument that provides improved depth accuracy. Additionally, a more sophisticated depth-denoising pipeline could further enhance measurement stability in complex environments. Similar computer vision frameworks have been implemented in industrial settings, such as measuring distances between structural components in the construction field [30] and performing deep learning-based visual inspection of gap spacing in high-precision equipment [31]. These examples demonstrate that the proposed RGB-based segmentation and depth-assisted measurement framework can be adapted for various industrial inspection and quality control applications.

ACKNOWLEDGMENTS

We extend our sincere gratitude to The U.S. Army Engineer Research and Development Center for their generous support and funding.

APPENDIX A DEPTH DENOISING PIPELINE

Let $D(x, y)$ denote the original recorded depth map, which is subject to various sources of noise. The denoising pipeline consists of the following steps:

A. MEDIAN BLUR FILTERING (OUTLIER REMOVAL)

A horizontal median filter with a kernel size of 1×5 is applied to suppress isolated noise while preserving structural edges:

$$\tilde{D}_0(x, y) = \mathcal{M}_{1 \times 5}[D](x, y)$$

B. GAP FILLING (ZERO-VALUE INTERPOLATION)

Missing depth values encoded as zeros are filled using the average of valid neighboring values:

$$\tilde{D}_1(x, y) = \begin{cases} \frac{1}{|\mathcal{N}(x, y)|} \sum_{(i, j) \in \mathcal{N}(x, y)} \tilde{D}_0(i, j), & \text{if } \tilde{D}_0(x, y) = 0 \\ \tilde{D}_0(x, y), & \text{otherwise} \end{cases}$$

where $\mathcal{N}(x, y)$ denotes the set of non-zero pixels in a local neighborhood around (x, y) .

C. BASELINE ESTIMATION (VEGETATION SUPPRESSION)

A strong vertical Gaussian blur with kernel size 1×401 is applied to create a smooth baseline depth map.

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad [32]$$

$$B(x, y) = \mathcal{G}_{1 \times 401}[\tilde{D}_1](x, y)$$

D. VEGETATION CORRECTION (DIFFERENCE THRESHOLDING)

Depth values significantly deviating from the baseline are replaced with the baseline:

$$\tilde{D}_2(x, y) = \begin{cases} B(x, y), & \text{if } |\tilde{D}_1(x, y) - B(x, y)| > T \\ \tilde{D}_1(x, y), & \text{otherwise} \end{cases}$$

where T is a predefined threshold.

E. PROGRESSIVE SMOOTHING (MULTI-SCALE GAUSSIAN BLUR)

Three consecutive vertical Gaussian filters with increasing kernel sizes are applied to smooth the depth map:

$$\tilde{D}_3(x, y) = \mathcal{G}_{1 \times 31}[\tilde{D}_2](x, y)$$

$$\tilde{D}_4(x, y) = \mathcal{G}_{1 \times 51}[\tilde{D}_3](x, y)$$

$$\tilde{D}_5(x, y) = \mathcal{G}_{1 \times 101}[\tilde{D}_4](x, y)$$

F. FINAL OUTPUT

The final depth map is denoted as:

$$\hat{D}(x, y) = \tilde{D}_5(x, y)$$

REFERENCES

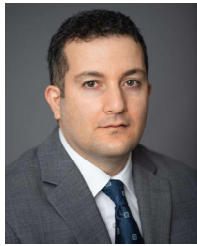
- [1] O. Galushko, Y. Kovalenko-Marchenkova, N. Chernova, A. Kasadzhik, and B. Gecbaia, "Impact of railway infrastructure on sustainable development of regions," in *E3S Web of Conferences*, vol. 558. EDP Sciences, 2024, p. 01035.
- [2] P. Bocz, N. Liegner, Á. Vinkó, and S. Fischer, "Investigation of the causes of railway track gauge narrowing," *Vehicles*, vol. 5, no. 3, pp. 949–977, 2023.
- [3] F. A. Prasetyo, S. A. Sunarto, Suprijanto, and L. Gunawan, "Railway track width gauge performance review: Evaluation of test results and their implications for safety improvement," *IPTEK Jurnal Nasional AMORI*, vol. 3, no. 1, pp. 40–48, 2024.
- [4] A. Moskal and E. Pastucha, "Track and gauge geometry measurements – the present and future," *Measurement Automation Monitoring*, vol. 62, no. 2, pp. 66–71, 2016.
- [5] Railroad Tools and Solutions, Inc., "Rolling track gauge, part no. rrt-4022-14," <https://rrtools.com/product/rolling-track-gauge/>, 2025, accessed: 2025-10-31.
- [6] L. Peng, H. Zhang, X. Li, and S. Zheng, "Inertial measurement system for track alignment inspection based on machine vision," in *Resilience and Sustainable Transportation Systems*, 2020, pp. 530–537.
- [7] J. Sresakoolchai and S. Kaewunruen, "Railway defect detection based on track geometry using supervised and unsupervised machine learning," *Structural health monitoring*, vol. 21, no. 4, pp. 1757–1767, 2022.
- [8] M. Ahmed, "Predicting track geometry using machine-learning methods," Master's thesis, University of Delaware, 2023.
- [9] K. Popov, R. De Bold, H. Chai, M. Forde, C. Ho, J. Hyslip, H. Kashani, R. Kelly, S. Hsu, and M. Rippin, "Data-driven track geometry fault localisation using unsupervised machine learning," *Construction and Building Materials*, vol. 377, p. 131141, 2023.
- [10] A. Pires, M. Viana, L. Scaramussa, G. Santos, P. Ramos, and A. Santos, "Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning," *Engineering Applications of Artificial Intelligence*, vol. 127, p. 107191, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0952197623013751>
- [11] K. R. Jerripothula, S. A. Ansari, and R. Nijhawan, "A vision-based solution for track misalignment detection," in *2021 34th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)*, 2021, pp. 271–277.
- [12] R. Tang, L. De Donato, N. Besinović, F. Flammini, R. M. Goverde, Z. Lin, R. Liu, T. Tang, V. Vittorini, and Z. Wang, "A literature review of artificial intelligence applications in railway systems," *Transportation Research Part C: Emerging Technologies*, vol. 140, p. 103679, 2022.
- [13] A. Kumar and S. P. Harsha, "A systematic literature review of defect detection in railways using machine vision-based inspection methods," *International Journal of Transportation Science and Technology*, vol. 18, pp. 207–226, 2025.
- [14] Intel Corporation, "Intel realsense depth camera d435," n.d. [Online]. Available: <https://www.intel.com/content/www/us/en/products/sku/128255/intel-realsense-depth-camera-d435/specifications.html>
- [15] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.
- [16] C. Yang, H. Huang, W. Chai, Z. Jiang, and J. Hwang, "Samurai: Adapting segment anything model for zero-shot visual tracking with motion-aware memory," 2024.
- [17] N. Ravi, V. Gabeur, Y. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, "Sam 2: Segment anything in images and videos," 2024. [Online]. Available: <https://arxiv.org/abs/2408.00714>
- [18] G. Jocher, "Ultralytics yolov5," 2020. [Online]. Available: <https://github.com/ultralytics/yolov5>
- [19] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics yolov8," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [20] C. Wang, I. Yeh, and H. Liao, "Yolov9: Learning what you want to learn using programmable gradient information," in *Computer Vision–ECCV 2024*, vol. 15089, 2024, pp. 1–21.
- [21] G. Jocher and J. Qiu, "Ultralytics yolo11," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [22] L. Qiao, M. Veltrup, and B. Mayer, "Comparison of mask r-cnn and yolov8-seg for improved monitoring of the pcb surface during laser cleaning," *Scientific Reports*, vol. 15, no. 1, p. 17185, 2025.
- [23] Y. Wu, Q. Han, Q. Jin, J. Li, and Y. Zhang, "Lca-yolov8-seg: an improved lightweight yolov8-seg for real-time pixel-level crack detection of dams and bridges," *Applied Sciences*, vol. 13, no. 19, p. 10583, 2023.
- [24] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W. Lo, P. Dollár, and R. Girshick, "Segment anything," 2023. [Online]. Available: <https://arxiv.org/abs/2304.02643>
- [25] Q. Zhou, C. Yan, Y. Zhu, X. Bai, and W. Liu, "Image labeling by multiple segmentation," in *2011 18th IEEE International Conference on Image Processing*. IEEE, 2011, pp. 3129–3132.
- [26] C. Yang, "Samurai: Segment anything model for tracking (code snippet)," https://github.com/yangchris11/samurai/blob/76ba195984892b0d1e3db5d9c9f90bb62175680a/sam2/sam2/sam2_video_predictor.py#L973-L981, 2024.
- [27] United States Department of Defense, "Ufc 4-860-03: Railroad design and evaluation," Unified Facilities Criteria (UFC), Tech. Rep. 4-860-03, 2008. [Online]. Available: https://www.wbdg.org/FFC/DOD/UFC/ufc_4_860_03_2008.pdf
- [28] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015*, vol. 9351, 11 2015, pp. 234–241.
- [29] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Computer Vision–ECCV 2018*, vol. 11211, 10 2018, pp. 833–851.
- [30] H. Perez and J. H. M. Tah, "Towards automated measurement of as-built components using computer vision," *Sensors*, vol. 23, no. 16, 2023.
- [31] X. Li, F. Li, H. Yang, and P. Wang, "Deep learning-enabled visual inspection of gap spacing in high-precision equipment: A comparative study," *Machines*, vol. 13, no. 2, 2025.
- [32] B. Jähne, *Digital image processing*, 6th ed. Springer, 2005.



VAN TRUNG LE is a Bachelor's student in Computer Science at Marshall University. His research interests include machine learning and data science, with a focus on applying these techniques to real-world applications.



HWAPYEONG SONG is a PhD student studying computer science in Engineering at Marshall University. Song received the B.S. in Computer Science and M.S. in Cybersecurity from Marshall University. Her research interests include computer vision, image processing, machine learning, software architecture, authentication algorithm, and vulnerability analysis. She has been involved in interdisciplinary research, including a notable project on machine learning applications in civil engineering and bioinformatics. Her current research focuses on developing advanced algorithms and models to address complex real-world problems across various domains.



HUSNU S. NARMAN is an Associate Professor in the Department of Computer Science at Marshall University, where he leads interdisciplinary research at the intersection of distributed computing and intelligent systems. Prior to joining Marshall, he completed a postdoctoral fellowship at Clemson University, following his Ph.D. in Computer Science from the University of Oklahoma. His work has resulted in over 70 peer-reviewed publications and has been supported by more than \$4 million in funding. His contributions to research, teaching, and service have been recognized with multiple honors, including the Weisberg Faculty Research Award, Weisberg Academy of Distinguished Teachers Award, Weisberg Service Award, and the Marshall University Distinguished Artists and Scholars Junior Category Award.



PINGPING ZHU is an Assistant Professor in the Department of Computer Science and Electrical Engineering at Marshall University. Prior to joining Marshall University, he was a Research Associate in the Sibley School of Mechanical and Aerospace Engineering at Cornell University. Before that, He worked as a Postdoctoral Associate in the Department of Mechanical Engineering and Materials Science at Duke University. Dr. Pingping Zhu received the Ph.D. and M.S. degrees in the Department of Electrical and Computer Engineering from the University of Florida in 2013 and 2010, respectively, and the M.S. degree from the Institute for Pattern Recognition and Artificial Intelligence and B.S. degree in Department of Electronics and Information Engineering in Huazhong University of Science and Technology, China, in 2008 and 2006, respectively. His research interests include signal processing, machine learning, artificial intelligence, intelligent control, and human behavior research.



AMMAR ALZARRAD is an Associate Professor in the Department of Civil Engineering at Marshall University. He graduated with dual bachelor's degrees in Civil Engineering and Business Administration from the University of South Alabama. He received his M.Sc. and Ph.D. in Civil Engineering from The University of Alabama. Before joining to Marshall University, he was an Assistant Professor in the Department of Civil Engineering and Construction at Bradley University. Prior to joining academia, Dr. Alzarrad was a Virtual Design and Construction (VDC) manager at an engineering design firm in Chicago, where he managed multi-million projects (i.e., Wrigley Field restoration and expansion project). Dr. Alzarrad is a PMP®, CPEM®, and the Director of The Engineering Management Graduate Program at Marshall University.



ABBY CISKO is a Research Mechanical Engineer in the Geotechnical and Structure's Laboratory at U.S. Army's Engineer Research and Development Center. She brings extensive expertise in railroad engineering, materials characterization, and large-scale infrastructure testing. She has multiple years of experience conducting microstructural and mechanical characterization of fatigue samples, as well as leading full-scale test section construction and controlled laboratory evaluations to assess infrastructure performance and degradation. Dr. Cisko has managed projects for the Federal Railroad Administration (FRA) and multiple DoD services, including the U.S. Navy, IMCOM, and U.S. Air Force. Her work emphasizes the performance of subgrade, ballast, and track structures under demanding operational conditions. In addition to her applied research, she actively collaborates with academic institutions to advance railroad science and develop innovative solutions for resilient military and civilian rail systems.

JEREMY BEASLEY is a seasoned railroad professional. He brings over 15 years of experience to the field, including 6 years as Railroad Track Evaluation Program Manager at ERDC and 14 years as an Army Certified Track Inspector. He spearheaded the modernization of field data collection and reporting, enhancing efficiency and accuracy. Holding a B.S. in Risk Management and a Geospatial Technologies Certification from Mississippi State University, his research focuses on sensor integration, inspection automation, and contingency operation support. Beasley is a published author, co-authoring numerous reports and journal articles related to subsurface evaluation, application development and emerging technologies.